AN ADMISSIBLE SHORT EXACT SEQUENCE IN A DG CATEGORY GIVES A TRIANGLE

THE STACKS PROJECT AUTHORS

Editorial note. The counted statement is the author's own lemma environment labelled lemma-get-triangle (source lines 3707-3717); the proof environment begins two lines below it and is its own proof, closing at source line 3755.

Editorial note. No \ref or \cite occurs anywhere inside the retained span, so the package declares neither a source pointer nor a bibliography entry; axioms (A) and (B) are hypotheses named in the statement and developed earlier in the same chapter, and the proof only uses the splitting it writes down.

Editorial note. The proof contains an Xy-pic diagram and eight numbered identities displayed in an enumerate environment; both are part of the retained source range and are reproduced as drawn, so no figure file is needed.

Editorial note. The retained preamble is the chapter's own minimal preamble (sources/stacks-dga/preamble.tex, 217 lines), a verbatim excerpt of the native Xy-pic setup, the native theorem declarations with the mathematical macros, and the project's own no-op reference environments; no macro was invented or redefined.

Editorial scope. One original result of the Stacks Project's Differential Graded Algebra chapter, counted once: an admissible short exact sequence in a dg category gives a triangle whose consecutive compositions vanish in the homotopy category.

Allowed standard prerequisites (editorial).

- Differential graded category. A category enriched in graded modules with a differential on each hom complex satisfying the Leibniz rule with respect to composition.. It is the ambient setting and supplies the Leibniz rule used throughout the eight identities (source lines 3709-3710, 3740-3741).
- Admissible short exact sequence in a dg category. A composable pair x -> y -> z for which there is a diagram splitting y as x direct sum z in the graded sense, with the accompanying structure maps.. The diagram with the maps a, pi, b, s and the splitting y isomorphic to x direct sum z is the starting point of the proof (source lines 3721-3730).
- Homotopy of morphisms of complexes. Two morphisms of complexes are homotopic when their difference is of the form d(h) for some graded map h, and homotopic morphisms are equal in the homotopy category.. The final two claims are exactly that a delta and delta b are homotopic to zero, which is what makes the compositions vanish in K(A) (source lines 3745-3755).

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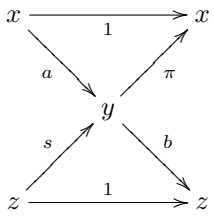
Lemma 0.1. *Let $\mathcal{A}$ be a differential graded category satisfying axioms (A) and (B). Given an admissible short exact sequence $x \to y \to z$ we obtain (see proof) a triangle*

$$x \to y \to z \to x[1]$$

in Comp($\mathcal{A}$) with the property that any two compositions in $z[-1] \to x \to y \to z \to x[1]$ are zero in $K(\mathcal{A})$.

Proof. Choose a diagram



giving the isomorphism of graded objects $y \cong x \oplus z$ as in the definition of an admissible short exact sequence. Here are some equations that hold in this situation

(1) $1 = \pi a$ and hence $\mathrm{d}(\pi)a = 0$,
(2) $1 = bs$ and hence $b\mathrm{d}(s) = 0$,
(3) $1 = a\pi + sb$ and hence $a\mathrm{d}(\pi) + \mathrm{d}(s)b = 0$,
(4) $\pi s = 0$ and hence $\mathrm{d}(\pi)s + \pi\mathrm{d}(s) = 0$,
(5) $\mathrm{d}(s) = a\pi\mathrm{d}(s)$ because $\mathrm{d}(s) = (a\pi + sb)\mathrm{d}(s)$ and $b\mathrm{d}(s) = 0$,
(6) $\mathrm{d}(\pi) = \mathrm{d}(\pi)sb$ because $\mathrm{d}(\pi) = \mathrm{d}(\pi)(a\pi + sb)$ and $\mathrm{d}(\pi)a = 0$,
(7) $\mathrm{d}(\pi\mathrm{d}(s)) = 0$ because if we postcompose it with the monomorphism a we get $\mathrm{d}(a\pi\mathrm{d}(s)) = \mathrm{d}(\mathrm{d}(s)) = 0$, and
(8) $\mathrm{d}(\mathrm{d}(\pi)s) = 0$ as by (4) it is the negative of $\mathrm{d}(\pi\mathrm{d}(s))$ which is 0 by (7).

We've used repeatedly that $\mathrm{d}(a) = 0$, $\mathrm{d}(b) = 0$, and that $\mathrm{d}(1) = 0$. By (7) we see that

$$\delta = \pi\mathrm{d}(s) = -\mathrm{d}(\pi)s : z \to x[1]$$

is a morphism in $\mathrm{Comp}(\mathcal{A})$. By (5) we see that the composition $a\delta = a\pi\mathrm{d}(s) = \mathrm{d}(s)$ is homotopic to zero. By (6) we see that the composition $\delta b = -\mathrm{d}(\pi)sb = \mathrm{d}(-\pi)$ is homotopic to zero. □